Q1

$$
\begin{aligned}
y &= \sin x \tan x \\
\frac{dy}{dx} &= \cos x \tan x + \sin x \sec^2 x \\
&= \sin x \left(1 + \sec^2 x\right)
\end{aligned}
$$

Q2

$$
\begin{aligned}
g(x) &= x^3 \sec x \\
g'(x) &= 3x^2 \sec x - (\sec x \tan x)\, x^3 \\
&= x^2 \sec x \,(3 - x \tan x)
\end{aligned}
$$

Q3

$$
\begin{aligned}
h(x) &= \sqrt{\cos\left(x^2 + 4x\right)} \\
h'(x) &= -\frac{1}{2}\left(\cos\left(x^2 + 4x\right)\right)^{\frac{1}{2}} \sin\left(x^2 + 4x\right)(2x + 4)
\end{aligned}
$$

Q4

$$
\begin{aligned}
g(x) &= e^{\sin x} \\
g'(x) &= e^{\sin x} \cdot \cos x
\end{aligned}
$$

Q5

$$
\begin{aligned}
y &= \tan x \\
\frac{dy}{dx} &= \sec^2 x
\end{aligned}
$$